

Distributed Databases

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Database System Concepts, 6th Ed.

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Distributed Database System

- A distributed database system consists of loosely coupled sites that share no physical component
- Database systems that run on each site are independent of each other
- Transactions may access data at one or more sites

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Homogeneous Distributed Databases

- In a homogeneous distributed database
  - All sites have identical software
  - Are aware of each other and agree to cooperate in processing user requests.
  - Each site surrenders part of its autonomy in terms of right to change schemas or software
  - Appears to user as a single system
- In a heterogeneous distributed database
  - Different sites may use different schemas and software
    - Difference in schema is a major problem for query processing
    - Difference in software is a major problem for transaction processing
  - Sites may not be aware of each other and may provide only limited facilities for cooperation in transaction processing

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Distributed Data Storage

- Assume relational data model
- Replication
  - System maintains multiple copies of data, stored in different sites, for faster retrieval and fault tolerance.
- Fragmentation
  - Relation is partitioned into several fragments stored in distinct sites
- Replication and fragmentation can be combined
  - Relation is partitioned into several fragments: system maintains several identical replicas of each such fragment.

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Data Replication

- A relation or fragment of a relation is **replicated** if it is stored redundantly in two or more sites.
- **Full replication** of a relation is the case where the relation is stored at all sites.
- Fully redundant databases are those in which every site contains a copy of the entire database.

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Data Replication (Cont.)

- Advantages of Replication
  - **Availability**: failure of site containing relation *r* does not result in unavailability of *r* is replicas exist.
  - **Parallelism**: queries on *r* may be processed by several nodes in parallel.
  - **Reduced data transfer**: relation *r* is available locally at each site containing a replica of *r*.
- Disadvantages of Replication
  - Increased cost of updates: each replica of relation *r* must be updated.
  - Increased complexity of concurrency control: concurrent updates to distinct replicas may lead to inconsistent data unless special concurrency control mechanisms are implemented.
    - One solution: choose one copy as **primary copy** and apply concurrency control operations on primary copy

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Data Fragmentation

- Division of relation r into fragments r_1, r_2, …, r_n which contain sufficient information to reconstruct relation r.
- **Horizontal fragmentation**: each tuple of r is assigned to one or more fragments
- **Vertical fragmentation**: the schema for relation r is split into several smaller schemas
  - All schemas must contain a common candidate key (or superkey) to ensure lossless join property.
  - A special attribute, the tuple-id attribute may be added to each schema to serve as a candidate key.

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Horizontal Fragmentation of *account* Relation

branch_name	*account_number*	*balance*
Hillside	A-305	500
Hillside	A-226	336
Hillside	A-155	62

$$account_1 = \sigma_{branch_name=\text{“Hillside”}}(account)$$

branch_name	*account_number*	*balance*
Valleyview	A-177	205
Valleyview	A-402	10000
Valleyview	A-408	1123
Valleyview	A-639	750

$$account_2 = \sigma_{branch_name=\text{“Valleyview”}}(account)$$

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Vertical Fragmentation of *employee_info* Relation

branch_name	*customer_name*	*tuple_id*
Hillside	Lowman	1
Hillside	Camp	2
Valleyview	Camp	3
Valleyview	Kahn	4
Hillside	Kahn	5
Valleyview	Kahn	6
Valleyview	Green	7

$deposit_1 = \Pi_{branch_name,\ customer_name,\ tuple_id}(employee_info)$

account_number	*balance*	*tuple_id*
A-305	500	1
A-226	336	2
A-177	205	3
A-402	10000	4
A-155	62	5
A-408	1123	6
A-639	750	7

$deposit_2 = \Pi_{account_number,\ balance,\ tuple_id}(employee_info)$

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Advantages of Fragmentation

- Horizontal:
  - allows parallel processing on fragments of a relation
  - allows a relation to be split so that tuples are located where they are most frequently accessed
- Vertical:
  - allows tuples to be split so that each part of the tuple is stored where it is most frequently accessed
  - tuple-id attribute allows efficient joining of vertical fragments
  - allows parallel processing on a relation
- Vertical and horizontal fragmentation can be mixed.
  - Fragments may be successively fragmented to an arbitrary depth.

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Query Transformation

- Translating algebraic queries on fragments.
  - It must be possible to construct relation *r* from its fragments
  - Replace relation *r* by the expression to construct relation *r* from its fragments
- Consider the horizontal fragmentation of the *account* relation into

  $account_1 = \sigma_{branch_name = \text{“Hillside”}}(account)$

  $account_2 = \sigma_{branch_name = \text{“Valleyview”}}(account)$
- The query $\sigma_{branch_name = \text{“Hillside”}}(account)$ becomes

  $\sigma_{branch_name = \text{“Hillside”}}(account_1 \cup account_2)$

  which is optimized into

  $\sigma_{branch_name = \text{“Hillside”}}(account_1) \cup \sigma_{branch_name = \text{“Hillside”}}(account_2)$

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Example Query (Cont.)

- Since $account_1$ has only tuples pertaining to the Hillside branch, we can eliminate the selection operation.
- Apply the definition of $account_2$ to obtain

  $\sigma_{branch_name = \text{“Hillside”}} (\sigma_{branch_name = \text{“Valleyview”}} (account))$

- This expression is the empty set regardless of the contents of the *account* relation.
- Final strategy is for the Hillside site to return $account_1$ as the result of the query.

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Heterogeneous Distributed Databases

- Many database applications require data from a variety of preexisting databases located in a heterogeneous collection of hardware and software platforms
- Data models may differ (hierarchical, relational, etc.)
- Transaction commit protocols may be incompatible
- Concurrency control may be based on different techniques (locking, timestamping, etc.)
- System-level details almost certainly are totally incompatible.
- A **multidatabase system** is a software layer on top of existing database systems, which is designed to manipulate information in heterogeneous databases
  - Creates an illusion of logical database integration without any physical database integration

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Query Processing

- Several issues in query processing in a heterogeneous database
- Schema translation
  - Write a **wrapper** for each data source to translate data to a global schema
  - Wrappers must also translate updates on global schema to updates on local schema
- Removal of duplicate information when sites have overlapping information
  - Decide which sites to execute query

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Mediator Systems

- **Mediator** systems are systems that integrate multiple heterogeneous data sources by providing an integrated global view, and providing query facilities on global view
  - Unlike full fledged multidatabase systems, mediators generally do not bother about transaction processing
  - But the terms mediator and multidatabase are sometimes used interchangeably
  - The term **virtual database** is also used to refer to mediator/multidatabase systems

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Directory Systems

- Typical kinds of directory information
  - Employee information such as name, id, email, phone, office addr, ..
  - Even personal information to be accessed from multiple places
    - e.g., Web browser bookmarks
- White pages
  - Entries organized by name or identifier
    - Meant for forward lookup to find more about an entry
- Yellow pages
  - Entries organized by properties
  - For reverse lookup to find entries matching specific requirements

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Directory Access Protocols

- Most commonly used directory access protocol:
  - LDAP (Lightweight Directory Access Protocol)
  - Simplified from earlier X.500 protocol
- Question: Why not use database protocols like ODBC/JDBC?
- Answer:
  - Simplified protocols for a limited type of data access, evolved parallel to ODBC/JDBC
  - Provide a nice hierarchical naming mechanism similar to file system directories
    - Data can be partitioned amongst multiple servers for different parts of the hierarchy, yet give a single view to user
      - E.g., different servers for Bell Labs Murray Hill and Bell Labs Bangalore
  - Directories may use databases as storage mechanism

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End of Course

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